Project Proposal

Visual Reasoning Through Animated Transformations

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1. Overview

Visual reasoning questions ask users to identify transformations (rotation, reflection, translation, scaling, etc..) between figures, usually shown as static images that must be mentally visualised. This project builds a responsive website that delivers hints either as text (static) or as step-by-step animated transformations, and studies how each affects cognitive load, performance, usability, sense of agency and understanding of transformation patterns.

2. Research Questions

- Do animated hints reduce cognitive load compared with static textual hints?
- Do animated hints improve accuracy and completion time?
- How does the sense of agency differ between modes, and does it relate to performance?
- Do animated hints help users learn transformation patterns and apply them to new questions?

3. System Design

Responsive layout. Mobile-first design for mobile (single column, large touch controls), iPad (side-by-side in landscape, stacked in portrait) and laptop (full layout with question, hint panel, timeline and controls). Scalable SVG/Canvas graphics.

Mode selection. The user taps one option on the start screen and stays in that workflow for the whole session:

Mode	Experience
Static	Figures shown as images, hints are textual only, revealed step by step.
Animation	Same questions; each hint is an animated transformation with playback controls.
Randomise	Each question is randomly (counterbalanced) given static or animated hints.

Animated hints with time windows. Each question is split into up to four transformations. Each transformation is one hint, shown within its own time window:

Hint	Time window	Content
Hint 1	T1	First transformation animates (e.g. translation)
Hint 2	T2	Second transformation animates (e.g. rotation)
Hint 3	T3	Third transformation animates (e.g. reflection)
Hint 4	T4	Fourth transformation animates (e.g. scaling)
Hint 5	Tn	Nth Transformation

- T1–Tn are per-hint time limits, starting values (about 15-30s) will be tuned in a pilot. Questions with fewer transformations use T1–Tn.
- Each hint shows its transformation name, step number and a timeline marking T1–T4.
- Controls: play, pause, replay, previous, next and speed for direct manipulation.

- Static and Animation modes present identical transformations in the same order, so they are comparable.

4. Measurement Framework

Construct	Method	Captured
Cognitive load	NASA-TLX; 1–9 effort rating per question, gaze/blink signals	Mental demand, effort, frustration
Performance	Automatic logging	Accuracy, time per question, hints used, replays/pauses, time per T1–T4
Usability	System Usability Scale (SUS)	Score 0–100 per mode and device after the activity.
Sense of Agency	Short Likert items (e.g. "I felt in control of the hints") plus control-usage logs	Perceived control
Pattern understanding	Transfer test on unseen questions, "Identify the transformation" prompts, gaze patterns	Rule learned and applied

5. Eye / Gaze Tracking

A webcam will record where users look during each hint to support cognitive-load and pattern-understanding analysis.

- **Tools:** WebGazer.js or MediaPipe Face Mesh (in-browser) are recommended.
- **Recorded:** gaze position over time aligned to T1–T4, dwell time and fixations on areas of interest (original, target and animated figure, hint label, controls).
- **Limits:** webcam accuracy is coarse, so will use large regions of interest and a calibration step. Full support on laptop, iPad on a stand, optional on mobile.
- **Privacy:** informed consent with opt-out, local processing, store only gaze coordinates (not video), anonymous IDs.

6. HCI Concepts

Cognitive load reduction · Animated transitions · Progressive disclosure · Recognition rather than recall · Direct manipulation and sense of agency · Responsive design · Usability evaluation (SUS) · Attention and gaze analysis.

7. Timeline

Timeframe	Milestone
Weeks 1–2	Collect, Synthesis and build 10 questions, define transformation sequences, text hints and T1–T4, finalise study design and measures.
Weeks 3–4	Build core system: figures, transformation logic, responsive layout, mode selection, logging.
Weeks 5–6	Timed animated hints, controls, Randomise mode, questionnaires, gaze tracking; integrate question set; pilot and tune.
Week 7	User testing and interviews across devices.
Week 8	Analyse results, refine prototype, prepare final presentation.

Note: if gaze tracking is unreliable on some devices, pattern understanding will rely on logs, the transfer test and interviews, with gaze reported as exploratory.

9. Expected Outcome

A functional, responsive prototype for mobile, iPad and laptop that lets users solve visual reasoning questions in Static, Animation or Randomise mode, with evidence on whether timed animated hints change cognitive load, performance, usability (SUS), sense of agency and understanding of transformation patterns.